The Pranava Institute

# FACT CHECKING AND VERIFICATION IN THE AGE OF AI

# ABOUT US

The Pranava Institute works at the **intersection of Emerging Technology, Public Policy, and Society from an India-first perspective.** We help organisations stay ahead of the curve on all things **tech policy** through research, strategic insight, and capacity-building. We believe in developing **emic approaches to technology and creating sustainable digital futures.**

# Our Work

The Pranava Institute
Center for Emerging Technology and Policy

PROJECT QUICK NOTE

**RESPONSIBLE AI DEPLOYMENT IN THE PUBLIC SECTOR**

SOCIO-TECHNICAL CONSIDERATIONS FOR THE PRE-DEPLOYMENT STAGE

Tech Ethics Lab

University of Notre Dame

**INDIA SEMICONDUCTOR REPORT: STATE OF PLAY**

Australian Government | Centre for Australia India Relations | Tech Policy Design Institute

A 360° Playbook for Policymakers

**Deepening Australia-India Cooperation on Critical Minerals**

April 2025

**BUILDING A RESILIENT REE SUPPLY CHAIN FOR INDIA**

# Our Work


**Feeling Automated: Ethical Development of Emotional & Interactive AI Systems**



## /YOUTH DIGITAL CULTURES LAB

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# Today's session:

- Why This Matters?
- What is AI? Does AI 'KNOW' things?
- When AI Generates False Information. Hallucination Vs Misinformation
- Deceptive design, Misinformation, AI
- AI as the Epistemic Authority. Epistemic hierarchies
- Whose Knowledge?
- How do you think about this in the AI age?
- How do you actually verify AI content?
- Can you design for verification?

# Let's Talk

Have you ever asked an AI tool a question and later found out the answer was wrong or you weren't sure if it was right?

What AI tools do you use on a daily basis? Do you ever check what they tell you?

# AI from four lenses:

## 01

### TECHNOLOGY

How do these tools actually work? What are their real capabilities and limits?

## 02

### SOCIETY

Who benefits from these tools, who is excluded, who bears the risks?

# AI from four lenses:

## 03

### DESIGN

Who designed these systems, for whom, and with what assumptions? What would AI tools designed for specific contexts look like?

## 04

### PUBLIC POLICY

What governance frameworks exist? What should they require? What can institutions do now, and what requires systemic change?

## WHAT IS GENERATIVE AI

Generative AI (GenAI) is a type of artificial intelligence that creates new content including text, images, video, audio, and software code by learning patterns from existing data.

## WHAT IS AGENTIC AI

Agentic AI acts as an autonomous system that plans, reasons, and executes multi step workflows using external tools. While Generative AI delivers outputs, Agentic AI delivers outcomes by operating with higher autonomy.

# DOES AI "KNOW" THINGS?

AI appears to:

- Answer questions
- Explain concepts
- Write essays
- Generate images
- Write code

However,

- It does not understand meaning
- It does not "know" facts

These systems are built using something called **Large Language Models.**

AI systems are based on
Large Language Models

These models are trained on books, websites, articles, conversations, basically all the text widely available online.

They learn patterns in language

# When AI Generates False Information

- Hallucination in AI
- Happens when data is incomplete and patterns are overgeneralised

AI tends to sound convincing because it is built for continuity and coherence

- Eg: Fake research citations; Incorrect statistics

# Hallucination vs Misinformation

**Hallucination:** AI generates false information as a consequence of how it works.
Examples: Fake research citations; fake statistics; non existent court cases

**Misinformation:** False information that spreads and influences people, regardless of whether the source intended harm
Examples: Deepfake videos on social media

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# Deceptive Design & Misinformation


Dark patterns attributes grouped by how they modify the user's choice architecture.

| Choice Architecture | Attribute | Description |
|---|---|---|
| Modifying the decision space | Asymmetric | Unequal burdens on choices available to the user |
| | Restrictive | Eliminate certain choices that should be available to users |
| | Disparate Treatment | Disadvantage and treat one group of users differently from another |
| | Covert | Hiding the influence mechanism from users |
| Manipulation the information flow | Deceptive | Induce false beliefs in users either through affirmative mis-statements, misleading statements, or omissions |
| | Information Hiding | Obscure or delay the presentation of necessary information to users |

**Information Promotion** — **Promoting engaging content regardless of the validity or safety of the information**

Facebook's Recommendation Algorithm

Mathur, A., Kshirsagar, M., & Mayer, J. (2021). What Makes a Dark Pattern... Dark? Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems, 1–18. https://doi.org/10.1145/3411764.3445610

Pekka Kallioniemi,. (2022). Facebook's Dark Pattern Design, Public Relations and Internal Work Culture. 5. 38-54. 10.34624/jdmi.v5i12.28378.

# Deceptive Design X AI


***Source:*** *Sridhar, S. (2026, February). Feeling Automated: Exploring Policy Frameworks for the Ethical Development of Social AI (The Pranava Institute). Available at: Feeling Automated*

***As AI powered tools and applications become ubiquitous, the generative nature of AI could lead to endlessly diverse forms of deceptive design.***

Large Language Models are trained on:

- Websites
- Books
- Articles
- Public data

Have you thought about where do these come from?

***Whose knowledge is this?***

# AI as the epistemic authority

- AI tools have increasingly become the **trusted sources of knowledge.**
- People rely on them to **know, explain, and decide what is true**, similar to how we once relied on **teachers, books, or experts.**
- This shift changes who we **trust** in the process of learning.

- What is knowledge?
- What is a knowledge system?
- How many knowledge systems exist in the world?
- Why does this matter to you in the age of AI?

# Is fast tracking the process of gaining knowledge same as gaining knowledge faster?

“three crucial intellectual capacities: the ability to think productively within uncertainty, self-regulated learning, and ‘meditative thinking’ as opposed to merely calculative approaches to knowledge”

The process itself is the point.

You have a problem, you need an answer, you find the most efficient path to get there.

Corbin, T., Liang, Y., Bearman, M., Fawns, T., Flenady, G., Formosa, P., McKnight, L., Reynolds, J., & Walton, J. (2024). Reading at university in the time of GenAI. Learning Letters, 3, 35–35. https://doi.org/10.59453/ll.v3.35

# Did the world always have a single system of knowledge?

# Who decides what counts as universal knowledge?

Why is western knowledge accepted as the default while others are considered local?

# Have you thought about..

**Coloniality created a hierarchy of knowledge:**

- Western ➔ scientific, rational, advanced
- Non-Western ➔ traditional, primitive, emotional

**This hierarchy still shapes:**

- university curricula
- research funding
- global rankings

# IT IS IMPORTANT TO ASK

- **What do I think is knowledge?**
- **From where am I taught to think?**
- **What are the methods I use?**
- **Do these shape my conclusions?**

# What should decolonial thinking look like

**Knowledge always comes from somewhere.**

**Decolonial options ask:**

- **Who is speaking?**
- **From which history?**
- **With what power?**

**Decolonial options invite us to think freely, from where we are, without asking permission from a single global authority of knowledge.**

# So how do you think about this in the AI age?

AI systems can show Algorithmic Bias.
They are shaped by the data they are trained on.

**Check:**

- **Can I trace the source?**
- **What is the source of this information?**
- **Can I verify it?**
- **What are alternate ways of looking at it?**
- **Could bias be present?**

# So how do you think about this in the AI age?

Traditional information systems:
- retrieve stored information
- can often be traced to a source

Large language models:
- predict the statistically most likely continuation
- optimize for linguistic plausibility
- do not internally verify whether statements are true

*‘Emergent facts arise from the interaction between training data, model architecture, and user prompts; although often plausible, they remain probabilistic, context-dependent, and epistemically opaque.’*

*Source:*
*Laurence Dierickx, Andreas L. Opdahl & Carl-Gustav Lindén. Strategic Simplicity Gets the Most: Evaluating Prompting Techniques in Non-Expert AI-Assisted Fact-Checking. International Journal of Human–Computer Interaction 0:0, pages 1-16.*

# How do you actually verify AI content?

**There's no formula to do this.**

Some strategies:

1. **Always ask for source**s with any AI generated output. Once you have them, verify these sources to check their authenticity.
2. Do not rely on AI for **thinking, questions its outputs.** See how they stand ground against your research and knowing.
3. If you **can not verify a claim, do not rely on it.**

# Can you design for verification?

**How would you think of designing for the following:**

- Asking for sources. Eg. Source trace button?
- Cross- check across sources
- Pause before trusting. Eg. Creating friction before sharing
- Know what’s AI-made.

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# Thank you!

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